Error-metric benchmark: classical (Simulink) vs. SNN (Nengo) controllers

60 s trajectory-tracking mission with yaw initial condition $\psi(0) = 0.08$ rad; 1 kHz grid

Channel	Controller	MAE	MSE	RMSE	IAE	ISE	ITAE	ITSE	max $\lvert e \rvert$	SSE
x (m)	Classical	3.93×10^{-3}	6.59×10^{-5}	8.12×10^{-3}	0.2357	3.95×10^{-3}	2.062	0.0261	0.0309	9.52×10^{-5}
	SNN	0.0315	1.05×10^{-3}	0.0323	1.890	0.0627	56.680	1.835	0.0703	-0.0313
y (m)	Classical	4.60×10^{-3}	1.68×10^{-4}	0.0130	0.2759	0.0101	5.751	0.2086	0.0602	1.07×10^{-4}
	SNN	0.0317	1.10×10^{-3}	0.0332	1.902	0.0660	56.008	1.868	0.0892	0.0294
$-z$ (m)	Classical	9.89×10^{-3}	3.55×10^{-4}	0.0188	0.5934	0.0213	7.884	0.1104	0.0874	-1.16×10^{-3}
	SNN	0.0355	2.28×10^{-3}	0.0477	2.133	0.1367	51.000	2.018	0.2227	0.0269
θ (rad)	Classical	8.31×10^{-5}	1.07×10^{-7}	3.26×10^{-4}	4.98×10^{-3}	6.38×10^{-6}	0.0158	8.72×10^{-6}	4.50×10^{-3}	1.48×10^{-10}
	SNN	3.27×10^{-3}	2.84×10^{-5}	5.33×10^{-3}	0.1962	1.71×10^{-3}	4.365	0.0206	0.0357	-7.81×10^{-4}
ϕ (rad)	Classical	3.08×10^{-4}	9.46×10^{-7}	9.73×10^{-4}	0.0185	5.68×10^{-5}	0.3615	1.12×10^{-3}	5.63×10^{-3}	-8.57×10^{-11}
	SNN	8.97×10^{-3}	1.45×10^{-4}	0.0120	0.5384	8.67×10^{-3}	15.073	0.2110	0.0762	-3.62×10^{-3}
ψ (rad)	Classical	8.25×10^{-4}	4.38×10^{-5}	6.62×10^{-3}	0.0494	2.63×10^{-3}	0.0223	7.21×10^{-4}	0.0800	4.46×10^{-17}
	SNN	3.65×10^{-3}	7.69×10^{-5}	8.77×10^{-3}	0.2187	4.61×10^{-3}	3.970	0.0128	0.0843	2.22×10^{-3}

Errors: $e =$ reference $-$ response (mission reference for x, y, $-z$; each controller's own desired angle for θ, ϕ; $\psi_d = 0$ for ψ).

MAE $= \overline{|e|}$; MSE $= \overline{e^2}$; RMSE $= \sqrt{\overline{e^2}}$; IAE $= \int |e|\,dt$; ISE $= \int e^2 dt$; ITAE $= \int t|e|\,dt$; ITSE $= \int t\,e^2 dt$; SSE $=$ mean error over the final second (signed).
Units: translational channels in m (integrals m·s, $\mathrm{m}^2\cdot\mathrm{s}$, m·$\mathrm{s}^2$, $\mathrm{m}^2\cdot\mathrm{s}^2$); attitude channels in rad.

Simulation conditions (identical for both controllers)

- Time step: both simulators advance at $\Delta t = 1$ ms (1 kHz grid). Nengo: nengo.Simulator(model) with the default dt = 0.001 s and sim.run(60) $\rightarrow$ 60,000 steps (script Quad_Z_vs_theta_vs_phi_F-3.py, lines 352--354); verified in the exported data, whose time column is a uniform 0.001 s grid ($t = 0.001, \ldots, 60.000$ s). Simulink: fixed-step discrete solver with FixedStep = pid_dt = 0.001 s (victor_pid_params.m) $\rightarrow$ 60,001 samples ($t = 0, \ldots, 60$ s).
- Mission: identical references --- $x_d = 1/(1+e^{-(t-5)})$; $y_d = 0$ for $t < 15$ s, then $1.5/(1+e^{-((t-15)-5)})$; $-z_d = 3/(1+e^{-(t-4)})$; $\psi_d = 0$, with initial yaw offset $\psi(0) = 0.08$ rad in both implementations.
- Architecture: same cascade --- PID in the outer translational loops (x, y, z), PD in the inner attitude loops (θ, ϕ, ψ); the classical attitude integral gains are set to zero to match the SNN structure exactly. Same error convention $e =$ reference $-$ response; each controller generates its own desired angles, $\theta_d = \arctan(u_x/g)$ and $\phi_d = \arctan(u_y \cos\theta / g)$.
- Numerical internals (the one intentional difference): within each 1 ms step, Simulink integrates the rigid-body dynamics with a fourth-order Runge--Kutta scheme, while Nengo advances the spiking populations with synaptic filtering ($\tau = 0.1$ s); the 1 kHz grid is where the outputs are compared.
- Data sources: classical = Quadri_Victor_combination_PDinner_psi0_timeseries.csv (exact Python port of the Simulink model, validated against the Simulink export to 5.8×10^{-15}); SNN = repository UAV_Nengo, branch Hexa-codex-test1, folder Nengo/PID_3subsistemas.